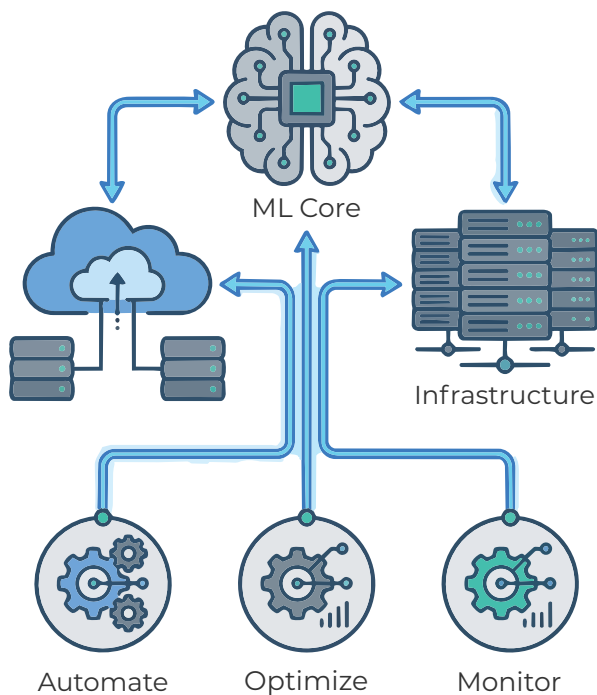


An ML-Driven Framework that Automates Infrastructure Management

The machine learning framework outlined here converts continuous infrastructure telemetry into weekly, automated resource governance decisions, closing the gap between observability and action using entirely open source tooling. It's particularly useful for infrastructure engineers, DevOps and SRE practitioners, database administrators, or platform engineering teams seeking to automate resource right-sizing at scale.



The consequences are well-documented. Industry benchmarks consistently report average server utilisation between 15 and 20 percent. For every five servers in a typical production fleet, the computational equivalent of four runs largely idle — provisioned against a worst case that rarely materialises, sustained by an operational culture that has no reliable mechanism to detect underutilisation and act on it.

The root cause is structural. The asymmetry of consequences in infrastructure operations creates a rational bias towards over-provisioning — under-provision and face immediate, visible incidents; over-provision and absorb invisible, silent cost. Without a system that learns from behavioural history and surfaces evidence-based recommendations, that bias persists indefinitely. So here's a framework that closes this gap — converting operational telemetry into weekly, automated resource governance decisions using a machine learning pipeline built entirely on open source tooling.

The reactive trap: Why static thresholds fail at scale

The dominant model for infrastructure resource management is reactive: define thresholds, fire alerts when thresholds are crossed, and respond to alerts. This model has a fundamental limitation — it can only respond to states, not trajectories.

Consider a PostgreSQL database whose connection utilisation climbs steadily over four weeks: 61%, 71%, 79%, 89%. At no point does an individual reading cross a critical threshold — so no alert fires. A human who reviews dashboards at weekly intervals would need to consciously track the trend across multiple reviews to detect it. At fleet scale — tens or hundreds of database instances — that vigilance is operationally impossible. The incident that follows is framed as unforeseeable, yet the signal existed in

Your databases are drowning in their own data. They emit thousands of metrics every minute — CPU curves, memory trends, connection rates, I/O pressure — and almost none of it informs the single decision that matters most: whether those databases are the right size. Modern infrastructure operations face a persistent paradox: the systems being managed generate continuous, high-fidelity telemetry, yet provisioning decisions — the most consequential operational choice affecting both cost and reliability — are made once, at deployment time, and rarely revisited.